

KITAKYUSHU AP, Japan

WMO: 478530

Lat: 33.846N

Lon: 131.035E

Elev: 18

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.8	35.7	17.6	13.4	37.4	19.8	14.9	39.2	30.1	38.3	27.8	38.8	15.1	290	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	89.9	78.0	88.0	78.1	86.3	77.6	80.0	85.7	79.4	85.1	78.8	84.5	10.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	149.5	83.4	77.3	142.2	83.1	76.9	140.3	83.0	43.8	86.1	43.0	85.2	42.2	84.6	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.1	23.1	20.5	30.3	93.9	3.0	1.7	28.2	95.1	26.4	96.1	24.8	97.1	22.6	98.4		
(5)			27.0	81.5	2.8	1.1	25.0	82.3	23.4	83.0	21.9	83.6	19.9	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	43.6	44.7	50.0	57.6	66.0	72.3	80.0	83.0	76.8	68.1	57.6	48.2	(6)
(7)		DBStd	14.03	4.22	4.60	4.72	4.37	3.65	3.30	3.95	3.24	3.87	4.11	5.21	4.87	(7)
(8)		HDD50	513	201	157	57	3	0	0	0	0	0	0	4	91	(8)
(9)		HDD65	2711	663	568	464	223	32	0	0	0	0	16	225	520	(9)
(10)		CDD50	5054	2	9	58	231	497	670	931	1023	803	560	233	37	(10)
(11)		CDD65	1777	0	0	0	1	64	220	466	558	353	111	4	0	(11)
(12)		CDH74	13871	0	0	1	5	96	607	4257	6356	2356	192	1	0	(12)
(13)		CDH80	4056	0	0	0	0	8	62	1239	2340	401	6	0	0	(13)
(14)	Wind	WSAvg	9.1	9.8	9.3	9.3	9.2	8.7	8.4	8.2	9.4	9.2	8.9	8.7	10.3	(14)
(15)	Precipitation	PrecAvg	76.8	2.4	3.2	4.8	5.7	6.6	14.5	12.5	7.8	8.8	4.7	3.5	2.4	(15)
(16)		PrecMax	122.1	7.1	6.2	13.4	10.1	12.0	27.2	27.2	15.3	20.0	14.2	7.7	5.0	(16)
(17)		PrecMin	46.0	0.7	1.1	1.8	1.6	2.2	2.8	2.8	2.1	1.9	0.6	0.9	0.2	(17)
(18)		PrecStd	15.0	1.5	1.3	2.2	2.1	2.5	6.1	7.0	4.1	5.2	3.5	1.8	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	60.6	67.8	73.1	80.6	84.4	91.8	93.5	87.9	80.6	71.3	62.4	(19)
(20)			MCWB	49.5	55.2	58.0	57.7	64.2	73.8	77.7	78.3	76.7	71.3	63.5	57.6	(20)
(21)		2%	DB	53.4	55.8	62.5	68.4	76.8	82.1	89.6	91.3	85.7	78.4	69.4	59.1	(21)
(22)			MCWB	47.4	50.6	54.1	57.2	62.6	72.6	77.2	78.4	75.4	70.7	62.7	54.0	(22)
(23)		5%	DB	51.6	53.6	59.1	66.2	73.7	79.0	87.6	89.3	84.0	75.6	67.6	57.2	(23)
(24)			MCWB	46.1	48.6	52.6	57.0	63.5	71.0	77.8	78.4	75.3	68.0	61.3	52.0	(24)
(25)		10%	DB	49.9	51.6	57.0	64.2	71.7	77.2	85.8	87.7	82.4	73.8	65.8	55.4	(25)
(26)			MCWB	45.2	47.0	51.5	56.3	63.1	70.3	77.4	78.4	74.6	66.0	60.0	50.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	56.2	59.6	62.8	69.4	76.4	80.9	81.5	79.5	74.7	67.0	59.5	(27)
(28)			MCDWB	55.4	60.2	64.8	67.1	73.3	81.4	85.6	86.7	83.8	77.9	69.4	61.4	(28)
(29)		2%	WB	49.4	52.3	56.0	60.9	67.9	74.7	79.9	80.3	78.1	72.2	64.9	55.4	(29)
(30)			MCDWB	51.1	55.1	59.8	65.0	71.4	78.4	85.0	86.4	82.7	76.7	67.6	58.3	(30)
(31)		5%	WB	47.5	50.0	54.1	59.6	66.8	73.6	79.1	79.8	76.9	70.0	62.8	53.3	(31)
(32)			MCDWB	50.0	52.3	58.1	63.7	70.7	77.5	84.5	85.9	82.1	74.6	65.9	56.2	(32)
(33)		10%	WB	45.4	47.8	52.3	58.4	65.3	72.4	78.3	79.2	75.8	67.6	60.2	51.5	(33)
(34)			MCDWB	48.8	50.8	56.3	62.5	69.8	76.0	83.8	85.2	80.9	72.6	64.4	54.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.8	10.1	10.3	10.2	7.9	7.8	8.0	7.9	8.6	8.6	8.1	(35)
(36)			MCDBR	10.4	11.5	13.6	13.9	13.5	11.0	10.7	10.1	8.9	9.6	9.7	9.7	(36)
(37)			MCWBR	7.6	8.6	9.0	7.8	7.2	5.6	4.4	4.4	4.7	5.6	7.0	8.0	(37)
(38)		5% WB	MCDWB	9.4	10.1	12.2	10.4	10.2	8.0	8.5	8.3	7.3	7.9	8.0	8.3	(38)
(39)			MCWBR	8.3	9.0	9.4	7.3	6.6	4.8	4.1	4.2	4.4	6.3	7.1	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.428	0.469	0.531	0.553	0.558	0.571	0.570	0.534	0.502	0.457	0.430	0.420	(40)	
(41)		taud	2.066	1.992	1.845	1.846	1.895	1.946	1.994	2.094	2.136	2.164	2.152	2.102	(41)	
(42)		Ebn at Noon	237	241	237	239	239	235	235	241	243	243	236	231	(42)	
(43)		Edh at Noon	42	50	62	65	63	60	56	50	46	42	38	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	687	914	1257	1525	1662	1427	1547	1624	1290	1088	808	636	(44)	
(45)		RadStd	62	87	89	132	188	161	231	176	133	113	74	43	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (31 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page